

Evaluation of Low-Cost Vibration Sensing Systems for Bearing Fault Detection

Interim Report and Project Plan

Author: Reuben Larsen¹

Supervisor: Pietro Borghesani

UNSW Sydney, School of Mechanical & Manufacturing Engineering

Abstract

Bearing failures are a leading cause of downtime in industrial mechanical systems, making early fault detection essential. While piezoelectric accelerometers are the established standard for vibration-based condition monitoring, their high cost limits widespread deployment. Low-cost MEMS accelerometers present a promising alternative, however the trade-off between cost and detection capability remains insufficiently characterised for early-stage faults. This work evaluates a low-cost MEMS-based measurement chain against a piezoelectric reference across varying outer race fault severities, aiming to quantify the loss in detection capability, identify the primary contributing factors, and investigate whether hardware constraints of the low-cost system can be exploited rather than mitigated.

Nomenclature

A	Amplitude	ADC	Analog-to-digital converter
f_n	Natural frequency of sensor	$BPFO$	Ball pass frequency of the outer race
f_r	Shaft rotational frequency	DAQ	Data acquisition system
f_s	Sampling frequency	FFT	Fast Fourier transform
k	Spring constant	$IEPE$	Integrated electronics piezoelectric
m	Seismic mass	$MEMS$	Micro-electromechanical systems
α	Contact angle of bearing	RMS	Root mean square
ω_r	Natural resonance frequency	SES	Squared envelope spectrum
		SNR	Signal-to-noise ratio

¹reuben.larsen@student.unsw.edu.au

1 Introduction

Industrial processes are predominantly driven by mechanical and electromechanical systems, of which a significant proportion consist of rotating machinery [7]. Within these systems, bearing failures are a leading cause of malfunction, accounting for approximately 50% of faults in electric motors [8]. Bearing defects should be detected at an early stage, as even small localised faults can degrade machine performance and progressively propagate into severe damage. If left undetected, these faults may result in catastrophic failure and secondary damage to surrounding components. Consequently, early fault detection is essential for reducing downtime, maintenance costs, and safety risks [22][6].

Vibration-based condition monitoring is widely adopted as the standard method for early bearing fault detection because of its effectiveness and convenience over other methods [13]. This approach utilises accelerometers, data acquisition systems, and signal processing techniques to identify characteristic vibration signatures associated with specific fault types. In particular, piezoelectric accelerometers are commonly used due to their high sensitivity, wide bandwidth, and ability to capture high-frequency fault-related content [16]. These systems enable the detection of characteristic fault frequencies generated by localised defects, forming the basis for vibration-based diagnostic techniques [22].

Despite their effectiveness, industrial-grade piezoelectric accelerometers and associated measurement systems are often costly, with individual sensors typically priced in the order of several hundred [4] to thousands of dollars for high grade industrial sensors [22]. In addition, such systems may require specialised signal conditioning, cabling, and additional data acquisition hardware, increasing both system complexity and deployment cost [16]. As a result, large-scale implementation across extensive industrial assets is often impractical, limiting the accessibility of continuous condition monitoring.

Low-cost micro-electromechanical systems (MEMS) accelerometers present a promising alternative, offering compact, lightweight, and inexpensive sensing solutions that can be integrated with low-cost analog-to-digital converters to form a complete low-cost measurement chain [22]. However, MEMS sensors typically exhibit lower sensitivity, higher noise floors, and reduced bandwidth compared to piezoelectric accelerometers, which may limit their ability to detect fault signatures, particularly in early-stage defects where signal amplitudes are low and often masked by noise [16]. Consequently, while prior studies have demonstrated the feasibility of MEMS-based condition monitoring systems [22], the trade-off between cost and detection performance remains insufficiently quantified. In particular, the minimum detectable fault severity and the scalability of such low-cost approaches relative to conventional piezoelectric systems are not well established in current literature.

The proposed system builds on the work of Sharma et al. [22], who demonstrated that a NucleoF401RE microcontroller paired with an ADXL1002 can successfully detect bearing faults, however only for a single fault configuration. This work extends that approach by systematically evaluating detection performance across four fault severity levels, directly quantifying the performance gap between the low-cost and piezoelectric systems as a function of spall size.

The study aims to quantify the loss in detection capability associated with low-cost hardware, identify the primary factors contributing to this performance degradation, such as noise, bandwidth limitations, and sampling constraints, and evaluate strategies to mitigate these limitations. Through this approach,

the work seeks to establish the practical limits of low-cost bearing fault detection and assess the viability of MEMS-based systems for scalable condition monitoring applications.

2 Literature Review

This literature review establishes the technical foundation required to evaluate the feasibility of low-cost condition monitoring using MEMS accelerometers. It is structured to progressively build the link between bearing fault physics, signal processing requirements, and measurement hardware for fault detection.

The review begins with the physical basis of bearing faults and their diagnostic indicators in Section 2.1, outlining fault types, characteristic frequencies, and the signal characteristics that any measurement system must capture. The signal processing techniques required to extract fault information from the measured signal are then reviewed in Section 2.2, covering detection challenges, statistical fault indicators, and the demodulation-based envelope analysis pipeline. Finally, the sensing and data acquisition chain is examined in Section 2.3, evaluating piezoelectric and MEMS accelerometers as reference and low-cost alternatives respectively, alongside the analog-to-digital conversion constraints that further govern system performance. The detection capabilities of each measurement approach are then combined in Section 2.3.5, identifying the minimum system requirements for reliable fault detection and the alternative pathways available to low-cost systems.

Together, these sections frame the central research question: whether a low-cost MEMS-based measurement chain can reliably detect early-stage bearing faults, and what factors govern any loss in detection capability relative to a conventional piezoelectric system.

2.1 Bearing Faults and Diagnostics

An understanding of bearing fault mechanics is fundamental to this work, as it establishes the physical basis for the vibration signals that the measurement system must detect. This section therefore introduces the key components of a rolling element bearing, the mechanisms by which faults develop and generate vibration, and the characteristic frequencies used as diagnostic indicators. This knowledge directly informs the signal processing and hardware requirements discussed in Sections 2.2 and 2.3 respectively.

Rolling element bearings enable relative motion between rotating and stationary components while supporting radial and axial loads. A typical deep-groove ball bearing consists of four primary components: the inner ring, outer ring, rolling elements (balls), and a cage that maintains ball spacing [8]. These components are illustrated in Fig. 1.

The key geometrical parameters governing bearing kinematics include the number of rolling elements (n_b), the ball diameter (B_D), the pitch diameter (P_D), and the contact angle (α) [21]. The contact angle is defined as the angle between the plane perpendicular to the bearing axis and the line connecting the points of contact between a rolling element and the inner and outer race-

ways, as illustrated in Section A-A Fig. 1. For a purely radial load, as is the case in this work, where the bearing is loaded in the horizontal direction by a hydraulic system [27], the contact angle is nominally zero, meaning the load is transmitted through the rolling elements perpendicular to the shaft axis. In contrast, axial or combined loading conditions produce a non-zero contact angle, which shifts the contact point and alters the effective pitch diameter seen by each rolling element. These parameters collectively define the motion of the rolling elements relative to the inner and outer races, and directly determine the characteristic vibration signatures generated when bearing components degrade [17].

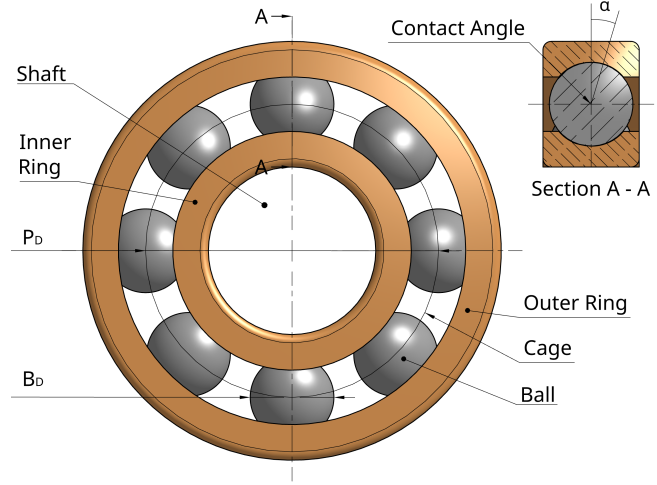


Figure 1: Ball bearing scheme, adapted from Magadán et al [8] and R.B. Randall et al [21]

2.1.1 Types of rolling element bearing faults

Bearing degradation commonly occurs due to excessive loading, poor lubrication, contamination, or shaft misalignment [8]. Over time these conditions can lead to localised surface damage, commonly referred to as a spall, as illustrated in Fig. 2.

The severity of a localised fault is commonly characterised by the size of the spall, typically expressed as its length in millimetres. Sub-millimetre defects represent the incipient fault regime where reliable detection remains challenging even for high-performance sensing systems. Spalls of approximately

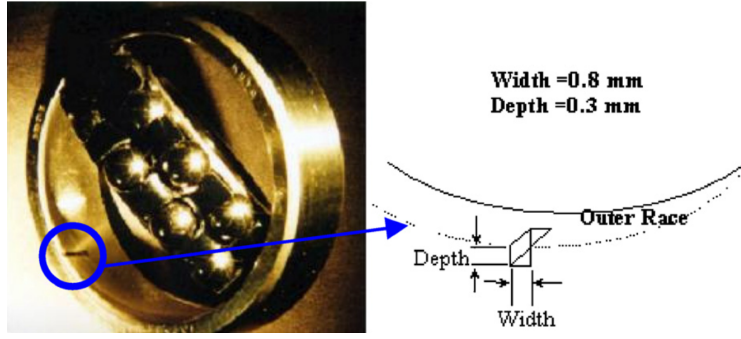


Figure 2: Example of a localised outer race spall defect [21].

1.0 mm represent the boundary of early-stage detection, with defects beyond this size considered a more developed fault condition [27] for the bearing size considered in this study.

The fault classification depends on the location of the defect [17][8]. Rolling element bearing faults are therefore typically categorised according to the affected component: outer race faults, inner race faults, rolling element (ball) faults, and cage faults. Each fault type generates vibration signatures at characteristic frequencies determined by the bearing geometry and shaft rotational speed [17]. These characteristic frequencies form the basis of vibration-based fault diagnostics. Table 1 summarises the primary fault types and their associated characteristic frequency.

Table 1: Common rolling element bearing fault types and their diagnostic characteristics [8]

Fault Type	Physical Description	Characteristic Frequency
Outer race	Localised defect on stationary outer raceway	BPFO
Inner race	Defect on rotating inner raceway	BPFI
Rolling element	Defect on ball or roller surface	BSF
Cage	Damage or looseness in the bearing cage	FTF

2.1.2 Characteristic Bearing Fault Frequencies

The characteristic frequencies represent the average rate at which impacts occur, and are determined by the bearing geometry and shaft speed. The frequency analysed for this thesis will be for outer race defects also known as the Ball Pass Frequency of the Outer race (BPFO)[17].

The BPFO is defined as

$$BPFO = \frac{n_b}{2} f_r \left(1 - \frac{B_D}{P_D} \cos \alpha \right) \quad (1)$$

where n_b is the number of rolling elements, f_r is the shaft rotational frequency, P_D is the pitch diameter, B_D is the ball diameter, and α is the contact angle of the bearing. [21][17].

The characteristic frequencies defined above form the primary diagnostic indicators of bearing faults,

and by identifying these frequencies in a measured vibration signal, the presence and type of fault can be determined. However, McFadden and Smith [11] established that fault-induced impulses are extremely short in duration, causing their energy to be distributed at a very low level over a wide range of frequencies, making bearing defects difficult to detect by conventional frequency spectrum analysis. This motivates the demodulation-based signal processing techniques discussed in Section 2.2.

2.2 Signal Processing for Bearing Fault Detection

Having established the physical basis of bearing fault signals in Section 2.1, this section reviews the signal processing techniques required to extract characteristic fault information from the measured vibration signal. It covers the detection challenges and signal characterisation in Section 2.2.1, and the demodulation-based envelope analysis pipeline in Section 2.2.2.

2.2.1 Detection Challenges and Signal Characteristics

As established in Section 2.1, fault-induced impulses are of extremely short duration, causing their energy to be distributed at a very low level over a wide range of frequencies, making bearing faults difficult to detect by conventional spectral analysis [11]. Furthermore, due to phenomena such as slip between rolling elements and raceways, the timing of successive impacts is not perfectly regular, meaning the signal is pseudo-cyclostationary rather than strictly periodic [15]. As a consequence, spectral lines at the characteristic fault frequencies are smeared across neighbouring frequencies rather than appearing as sharp discrete peaks. In practical machines, the fault signal must also compete with background vibration from other rotating components such as gears and shafts, further masking the fault-related content [11][15]. Fortunately, as noted by Randall and Antoni [15], the fault impacts excite structural resonances at much higher frequencies than the background vibration, concentrating some of the fault energy into a narrow band that is more readily detected, forming the physical basis of envelope analysis discussed in Section 2.2.2.

Prior to envelope analysis, statistical features such as RMS and kurtosis provide a simple means of characterising the overall signal condition [22][15] and are used in this work as performance metrics alongside the envelope analysis pipeline. However, statistical features characterise the signal globally and cannot identify the fault type or location. Kurtosis can be extended to the frequency domain as spectral kurtosis, enabling localisation of impulsive energy to specific frequency bands [23], forming the basis of the demodulation pipeline discussed in Section 2.2.2.

2.2.2 Demodulation-Based Fault Detection

McFadden and Smith [10] demonstrated that the vibration produced by a bearing fault can be modelled as a series of impulse responses of the bearing structure, where the amplitude of each response is modulated by the characteristic fault rate and bearing load distribution. The fault information is therefore contained in the modulation of a high-frequency carrier rather than in the carrier frequency itself. As noted by Randall and Antoni [15], envelope analysis was developed specifically to shift the diagnostic information from the high range of resonant carrier frequencies down to the much lower range of fault frequencies, where they can be resolved with good spectral resolution. This is achieved through a three-step pipeline: carrier band selection, bandpass filtering, and envelope extraction, each discussed below.

Firstly, the carrier band corresponds to a structural resonance of the bearing or surrounding machine structure, which is excited by the impulsive fault contacts and amplifies the fault-related signal content [15]. Since these structural resonances are not always known a priori, band selection methods such as the kurtogram are commonly employed to identify them. Some established diagnostic methods, such as the Shock Pulse Method and Spike Energy, take a different approach by deliberately exploiting the sensor's own resonance frequency f_n as a well-defined carrier band [15]. However, it has been noted that when the transducer resonance coincides with other excitation sources, such as turbulence or cavitation, false readings may result, and careful selection of the demodulation band is therefore recommended [15]. For low-cost MEMS sensors with significantly lower f_n , this carrier band shifts to much lower frequencies, with direct implications for fault detectability later discussed in Section 2.3.2.

Where the carrier frequency is not known, spectral kurtosis can be used to identify frequency bands containing impulsive fault-related energy. This is commonly visualised using a kurtogram, which maps spectral kurtosis across a range of centre frequencies and bandwidths, with regions of high kurtosis indicating bands dominated by impulsive behaviour associated with bearing faults [21][23]. The optimal band is selected as the region of maximum kurtosis, as illustrated in Fig. 3, where the 10–13 kHz band is identified as optimal for this example. It should be noted however that the kurtogram is a blind method, meaning it requires no prior knowledge of fault frequencies but can be misled by impulsive noise unrelated to the fault, and more targeted band selection methods exist where fault frequencies are known [23].

Once the optimal band is identified, a bandpass filter is applied to isolate the carrier, suppressing out-of-band noise and enhancing the fault-related signal component [15]. The filtered signal is then demodulated using the Hilbert transform to extract its amplitude envelope, effectively converting the

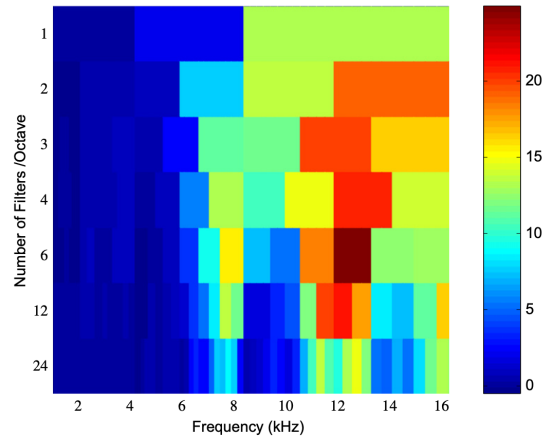


Figure 3: Example kurtogram showing the distribution of spectral kurtosis across frequency bands [21].

high-frequency carrier into a lower-frequency signal containing the modulation at the characteristic fault rate. The squared envelope spectrum (SES) is then computed, in which periodic impacts caused by bearing defects manifest as distinct peaks at BPFO and their harmonics [23][15], as illustrated in Fig. 4. These features are often not distinguishable in the raw frequency spectrum, particularly in the presence of noise or when fault amplitudes are low.

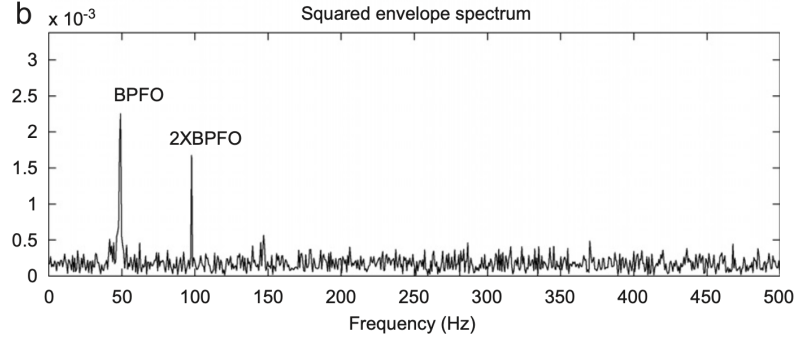


Figure 4: Envelope spectrum showing characteristic fault frequencies (BPFO and harmonics) [20].

The effectiveness of the demodulation pipeline therefore depends critically on the sensor bandwidth being sufficient to capture the carrier band, the ADC sampling rate being adequate to resolve the high-frequency content, and sufficient signal-to-noise ratio to distinguish fault-related peaks from background noise. These hardware requirements are examined in Section 2.3.

2.3 Sensors and Measurement Chain

Having established the nature of bearing fault signals and the signal processing techniques required to extract fault information in Section 2.2, this section reviews the measurement hardware required to capture them, as the effectiveness of fault detection depends on the combined performance of the sensing and data acquisition chain [16][24]. It first examines piezoelectric accelerometers and the properties that make them the reference standard for vibration-based fault detection in Section 2.3.1, before introducing MEMS accelerometers as an increasingly attractive low-cost alternative and the limitations that arise from their design in Section 2.3.2. The constraints imposed by analog-to-digital conversion and data acquisition hardware are then considered in Section 2.3.3, before the performance of these measurement chains is compared and their fault detection capabilities evaluated in Sections 2.3.4 and 2.3.5.

2.3.1 Piezoelectric Accelerometers

Piezoelectric accelerometers are the reference standard for vibration-based condition monitoring [22][1], and understanding the physical basis of their performance establishes the benchmark against which low-cost alternatives are evaluated in this work. The PCB 352C33, adopted as the reference sensor in this work, employs a ceramic shear design in which the piezoelectric element and seismic mass are configured as hollow cylinders mounted around a central stud, as illustrated in Fig. 5. Under base excitation, the inertia of the seismic mass exerts a shearing force on the piezoelectric ceramic, generating an electrical charge directly proportional to the applied acceleration [5][18]. Compared to compression mode designs, the shear configuration offers superior thermal stability as the pyroelectric effect is weakly pronounced [18].

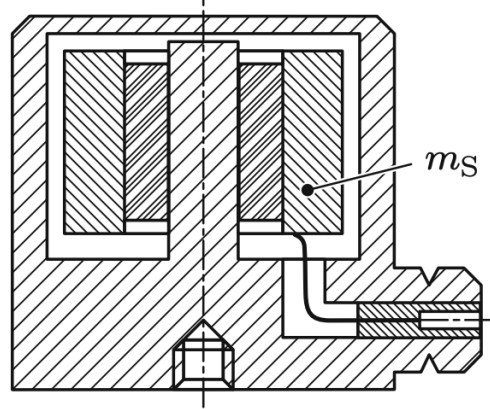


Figure 5: Shear mode piezoelectric accelerometer [18].

The key performance characteristics of piezoelectric accelerometers arise directly from the high stiffness of the sensing element. Modelling the sensor as a mass-spring system, the high stiffness relative to the seismic mass results in a natural frequency f_n typically in the range of tens to hundreds of kHz [5]. The usable frequency range is conventionally defined as the band within which the amplitude response remains within 5%, extending to approximately one-fifth of f_n [5], typically spanning from a few Hz to 15–30 kHz depending on the sensor. This wide bandwidth enables accurate capture of the high-frequency structural resonance excited by bearing fault impacts without significant attenuation or distortion [5].

A further consequence of the high stiffness is that the seismic mass undergoes negligible displacement under excitation relative to a compliant system and it is the strain within the piezoelectric element that generates the output signal [5]. This makes piezoelectric sensors inherently well suited to capturing transient impulsive events, such as the high-frequency bursts generated by localised bearing defects discussed in Section 2.1.

Industrial piezoelectric accelerometers typically exhibit sensitivities of ~ 100 mV/g, measurement ranges of ± 100 g, and bandwidths of 2.5 Hz–30 kHz, at a cost ranging from several hundred to several thousand dollars [22]. The PCB 352C33 adopted in this work represents a practical entry-level unit, offering a sensitivity of 100 mV/g, a measurement range of ± 50 g, and a frequency range of 0.5–10 kHz at a cost of approximately \$600 [14]. This cost limitation motivates the investigation of MEMS-based alternatives, discussed in Section 2.3.2.

2.3.2 MEMS Accelerometers

Capacitive MEMS accelerometers offer a low-cost and compact alternative for vibration measurement, and have seen increasing adoption in condition monitoring applications [16][1]. Rather than relying on strain within a stiff piezoelectric element, capacitive MEMS accelerometers measure acceleration through the displacement of a seismic mass. A seismic mass is a small silicon mass suspended by compliant beams that moves relative to the substrate under acceleration, with this displacement measured as a change in capacitance between the seismic mass and fixed electrode plates, as illustrated in Fig. 6 [26][1]. The size, cost, and ease of integration of these sensors make them particularly attractive for large-scale condition monitoring deployment [22][9].

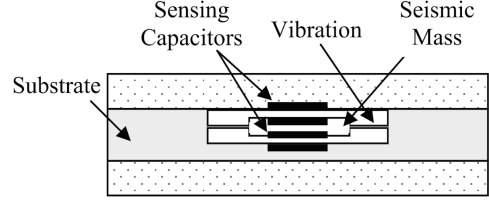


Figure 6: Typical capacitive MEMS accelerometer structure [1].

The performance characteristics of capacitive MEMS accelerometers arise directly from the compliant structure used to suspend the seismic mass, as the suspension beams result in a lower spring constant relative to the seismic mass and therefore a lower natural frequency f_n compared to piezoelectric sensors [26]. The flat response region extends to approximately one-fifth of f_n [5]. For low-cost MEMS sensors this may be as low as a few hundred Hz, well below the structural resonance frequencies typically excited by bearing faults and limiting the carrier band available for demodulation-based fault detection, with direct implications for the effectiveness of envelope analysis, as established in Section 2.2 [15].

A further consequence of the compliant structure is an elevated noise floor relative to piezoelectric sensors. The primary mechanical noise source in MEMS accelerometers is Brownian motion of the seismic mass and its suspension, which manifests as random fluctuations in the output signal [26]. This noise is inversely related to both the seismic mass and the mechanical quality factor of the structure [26], meaning that the small seismic masses characteristic of low-cost MEMS sensors result in an elevated noise floor that reduces the ability to detect low-amplitude fault signatures, particularly in early-stage defects [16].

High-performance MEMS sensors such as the ADXL1005 and ADXL1002 navigate these limitations through a small, stiff seismic mass optimised for high-frequency response, achieving bandwidths of 23 kHz and 21 kHz respectively at a cost of approximately \$55 and \$40 [3][2]. Notably, these bandwidths exceed that of the PCB 352C33 piezoelectric reference sensor adopted in this work, suggesting that for these MEMS candidates the performance comparison is governed primarily by noise floor and ADC sampling constraints rather than sensor bandwidth. However, even where a MEMS sensor achieves sufficient bandwidth, the ability to capture fault signatures is further constrained by the analog-to-digital conversion and data acquisition hardware, the performance of which is examined in the following section.

2.3.3 Analog to Digital Converters and Data Acquisition

The analog-to-digital converter (ADC) forms a critical component of the vibration measurement chain, governing the conversion of the continuous analog sensor output into a discrete digital signal for processing. Even when a sensor has sufficient bandwidth and noise performance, a poorly matched ADC can introduce limitations that bottleneck the overall system performance [24]. A representative low-cost data acquisition system integrating an ADC with a MEMS sensor is shown in Fig. 7, illustrating the key components of such a measurement chain [22].

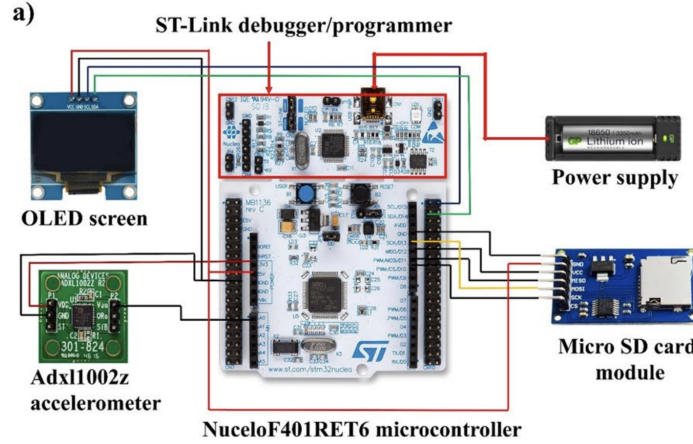


Figure 7: MEMS-based data acquisition system using an ADXL1002 accelerometer and NucleoF401RE microcontroller [22].

The two key ADC parameters governing measurement quality are sampling frequency and resolution. Sampling frequency determines the highest frequency that can be captured and, by the Nyquist criterion, the sampling rate must be at least twice the maximum frequency of interest [22]. Bearing structural resonances are typically excited in the range of 5–20 kHz [21], with values of 50 kHz commonly cited to ensure adequate capture of high-frequency fault signatures [24][27]. ADC resolution, expressed in bits, determines the number of discrete amplitude levels available to represent the signal, with a 12-bit ADC providing $2^{12} = 4096$ levels compared to over 16 million for a 24-bit device, directly affecting the ability to resolve low-amplitude fault signatures against background noise.

Specialised vibration data acquisition systems such as the National Instruments NI-9234 are designed specifically for this application, offering 24-bit resolution, a maximum sampling rate of 51.2 kS/s per channel, a built-in Delta-Sigma ADC architecture with integrated analog prefiltering and anti-aliasing, and native support for IEPE piezoelectric sensors [12], at a cost of \$3,717. In contrast, low-cost platforms such as the NucleoF401RE offer 12-bit ADC resolution at sampling rates around 5 kHz at a cost of \$20 [22], but lack dedicated signal conditioning, anti-aliasing filters, and the resolution required to match the dynamic range of high-performance sensors. Anti-aliasing filters, present in dedicated data acquisition systems such as the NI-9234, remove high-frequency content above the Nyquist frequency prior to sampling, preventing spurious aliased components from appearing in the digitised signal [15]. Whether the absence of such filtering is a fundamental limitation for vibration analysis or can in fact be exploited is discussed later in this section.

Sharma et al. further adopted the criterion that the sampling frequency must exceed 2.56 times the maximum frequency of interest to completely eliminate the possibility of aliasing effects [22]. Sharma’s study operated within a measurement bandwidth of 0–2500 Hz, with a sampling rate of 5 kHz, and successfully demonstrated envelope analysis at this frequency range. However, targeting the higher structural resonance bands of 5–20 kHz typically observed in industrial bearing applications [24][21] requires applying the same criterion to yield sampling rates requiring ~ 50 kHz. Sharma further noted that SPI communication delays, UART bridge delays, and SD card write times all reduce the actual achieved sampling frequency below the theoretical maximum [22], as modelled by:

$$F_s = \frac{1}{T_s + T_{SPI} + T_{UART} + T_{SD}} \quad (2)$$

where F_s is the actual sampling frequency, T_s is the sampling period, T_{SPI} is the SPI communication delay, T_{UART} is the UART bridge delay, and T_{SD} is the data storage delay [22]. This represents a practical constraint that must be carefully considered when targeting higher sampling rates, as the cumulative effect of these delays may prevent the system from achieving the ~ 50 kHz required for high-frequency envelope analysis.

Where the NucleoF401RE sampling rate falls below the Nyquist frequency of a high-frequency structural resonance band, frequency content from that band will alias into the measurable range rather than being rejected. Unlike in conventional systems where anti-aliasing filters prevent this, the absence of such filtering in low-cost platforms means aliased components are present in the digitised signal. Critically, since the characteristic fault frequencies are encoded in the amplitude modulation of the carrier rather than the carrier frequency itself, envelope analysis performed on an aliased carrier band can still recover BPFO and its harmonics intact. This raises the possibility that a low-cost system operating well below the Nyquist criterion for high-frequency content may nonetheless achieve reliable fault detection through aliased envelope analysis, without requiring dedicated anti-aliasing hardware or high sampling rates. This pathway is identified as a key experimental sub-question for Thesis B and is reflected in the research aims of Section 3.2.

The sensing and data acquisition characteristics discussed above directly govern the performance of the measurement chain, compared in the following section.

2.3.4 Performance Comparison

A comparison is made between piezoelectric and MEMS accelerometers, along with representative data acquisition systems, to evaluate their suitability for bearing fault detection. The analysis focuses on key parameters including bandwidth, noise, sensitivity, range, sampling frequency, and ADC resolution, which directly influence the ability to capture fault-related vibration signals. Sensor sensitivity, expressed in mV/g, is the ratio of its electrical output to its mechanical input [1], however its effectiveness is inherently tied to the sensor noise floor, as it is the ratio of signal to noise that ultimately governs fault detectability. Sensor range specifies the maximum acceleration that can be measured [1], and must be sufficient to accommodate the dynamic range of the system. A general comparison between piezoelectric and MEMS accelerometers is summarised in Table 2, with device-level

specifications provided in Table 3.

Table 2: Generalised Comparison of piezoelectric and MEMS accelerometers for vibration-based fault detection.

Characteristic	Piezoelectric	MEMS
Bandwidth	Wide (Generally up to 10–30+ kHz)	Sensor dependent (up to ~ 23 kHz)
Noise floor	Low	Higher
Sensitivity	High	Moderate (sensor dependent)
Size	Larger	Compact
Cost	High ($\sim \$600$ – $\$3000$ +))	Low ($\sim \$5$ – $\$55$)
Fault detectability	Established	Under investigation

Table 3: Specification of accelerometers used in condition monitoring, adapted from [22] with additional comparable sensors compiled from manufacturer datasheets.

Sensor	Bandwidth (kHz)	Sensitivity (mV/g)	Noise ($\mu\text{g}/\sqrt{\text{Hz}}$)	Range (g)	Cost (USD)
Piezoelectric industrial sensor used by [22]	2.5–30	100	~ 10	± 100	~ 3000
PCB 352C33 (Piezoelectric, reference sensor) [14]	0.5–10	100	~ 15	± 50	~ 600
ADXL1005 (MEMS) [3]	23	20	~ 25	± 100	~ 55
ADXL1002 (MEMS) [2]	21	40	~ 25	± 50	~ 40
ADXL335 (MEMS) [22]	0.55	330	~ 300	± 3	~ 5

The PCB 352C33 piezoelectric accelerometer serves as the reference sensor in this work, offering a bandwidth of 0.5–10 kHz, sensitivity of 100 mV/g, and a noise floor of approximately $15 \mu\text{g}/\sqrt{\text{Hz}}$ at a cost of approximately \$600 [14]. Of the MEMS candidates considered, the ADXL1005 and ADXL1002 demonstrate competitive performance at significantly lower cost, with bandwidths of 23 kHz and 21 kHz respectively, notably exceeding that of the PCB 352C33, suggesting that the performance comparison will be governed primarily by noise floor and ADC sampling constraints rather than bandwidth. In contrast, the ADXL335, despite its high sensitivity, exhibits a bandwidth of only 0.55 kHz, presenting a fundamental constraint for capturing high-frequency fault signatures, as discussed in Section 2.2.2.

However, sensor performance alone does not determine system capability, and the ADC must also be appropriately matched to avoid bottlenecking the measurement chain. Sharma et al. [22] demonstrated that the NucleoF401RE provides sufficient sampling rate and resolution for low-frequency envelope analysis, however whether this extends to higher frequency approaches is examined in Section 2.3.5. A comparison of the NucleoF401RE ADC against the NI-9234 specialised vibration DAQ is summarised in Table 4.

Table 4: Comparison of data acquisition systems for bearing fault detection, compiled from manufacturer datasheets [12][25].

System	ADC Resolution	Max Sampling Rate	Anti-aliasing	Cost (USD)
NI-9234 [12]	24-bit	51.2 kS/s	Built-in	\$3,717
NucleoF401RE [25]	12-bit	~ 5 kS/s	None	\$20

2.3.5 Detection Capabilities

Having characterised the sensing and data acquisition hardware, this section examines the minimum system requirements for reliable bearing fault detection, and identifies the detection pathways available to low-cost systems given their hardware constraints.

Conventional envelope analysis targets structural resonances in the 5–20 kHz range [24][21], placing significant demands on sensor bandwidth, sampling rate, and SNR that low-cost MEMS and ADCs may struggle to meet. However, two hardware constraints of low-cost systems can be exploited for fault detection rather than treated as limitations. First, the sensor’s own structural resonance can be exploited as a naturally available carrier band, an approach used by the Shock Pulse Method and Spike Energy [15] and discussed in Section 2.2.2, requiring only that the sensor bandwidth captures its own resonance frequency, significantly relaxing both the bandwidth and sampling requirements. Second, as established in Section 2.3.3, where the ADC sampling rate falls below the Nyquist frequency of a high-frequency resonance band, that content aliases into the measurable range, and since the fault frequencies are encoded in the amplitude modulation of the carrier signal, envelope analysis on an aliased carrier can still, in theory, recover BPFO intact. The implications of these three approaches for measurement system requirements are summarised in Table 5.

Table 5: Measurement system requirements for bearing fault detection under three envelope analysis approaches.

Parameter	Conventional High-Frequency	Sensor Resonance	Aliased Envelope
Sensor bandwidth	>21 kHz to capture 5–20 kHz band [24]	Sufficient to capture sensor f_n (normally low)	Potentially Any
Sampling frequency	>50 kHz based on 2.56 $\times$ rule [22]	~ 5 –10 kHz sufficient	Low ~ 5 kHz [22]
Carrier frequency	5–20 kHz structural resonance	Sensor f_n	Aliased high-frequency band
Cost implication	High	Low	Low

Both alternative pathways are identified as experimental sub-questions for Thesis B and are reflected in the research aims of Section 3.2.

3 Research Question and Project Plan

This section presents the research question and hypothesis derived from the literature review, defines the aims of the work, and outlines the experimental methodology and resources required to address them.

3.1 Research Question and Hypothesis

The literature review has established that low-cost MEMS accelerometers paired with low-cost ADCs offer a promising alternative to conventional piezoelectric sensors for vibration-based bearing fault detection, however the trade-off between cost and detection capability across the full measurement chain remains insufficiently characterised, particularly for early-stage faults requiring high-frequency envelope analysis. This work therefore addresses the following research question:

Can a low-cost vibration measurement chain based on a MEMS accelerometer and low-cost ADC reliably detect bearing fault signatures compared to a conventional piezoelectric accelerometer, and what are the primary factors governing any loss in detection capability?

It is hypothesised that a carefully selected low-cost MEMS measurement chain can achieve reliable bearing fault detection for sufficiently severe faults, but that detection capability will degrade relative to a piezoelectric reference system as fault severity decreases, with bandwidth limitations, noise floor, and ADC sampling constraints identified as the primary contributing factors.

3.2 Aims

The following aims form the basis of this research:

1. Quantify the loss in fault detection capability associated with low-cost MEMS hardware relative to a piezoelectric reference system across a range of fault severities.
2. Identify the primary factors contributing to performance degradation, including sensor bandwidth, noise floor, and ADC sampling constraints.
3. Investigate whether hardware constraints in the low-cost measurement chain can be exploited for fault detection, specifically the MEMS sensor's own structural resonance as a demodulation carrier band, and envelope analysis of an aliased signal from a high-frequency band arising from the low ADC sampling rate.

To address these aims, the experimental programme builds on the work of Sharma et al. [22], who demonstrated that a low-cost MEMS measurement chain can detect bearing faults for a single fault configuration. This present work extends this by evaluating detection capability across four fault severity levels and by investigating the unique hardware characteristics of the low-cost system, as detailed in Section 3.3.

3.3 Experimental Methodology

Experiments will be conducted on the UNSW bearing test rig at Willis Annexe J18, using a Nachi 6205-2NSE9 ball bearing. All tests will use outer race faults at a constant radial load of 10.5 kN and shaft speed of 20 Hz, with circular defects seeded using a milling machine. This rig and bearing configuration is identical to that used by Zhang et al. [27], providing a well-characterised experimental basis with known signal-to-noise characteristics. The planned tests follow the outline in Table 6.

Table 6: Planned experimental test matrix.

Test	Defect Position	Seeded Defect Size (mm)	Radial Load (kN)	Shaft Speed (Hz)
1	Outer race	0.25	10.5	20
2	Outer race	0.50	10.5	20
3	Outer race	1.00	10.5	20
4	Outer race	2.00	10.5	20

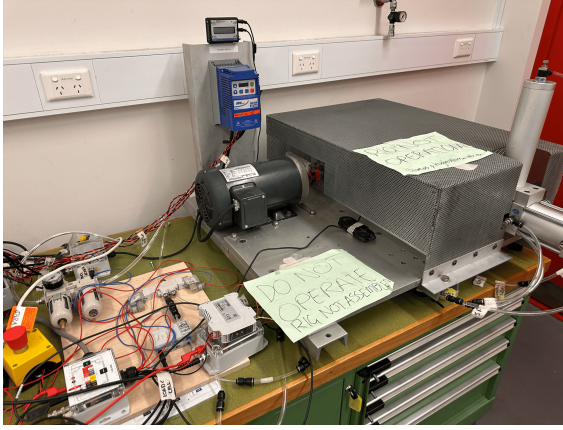
The 1.0 mm and 0.5 mm defect sizes correspond directly to those used by Zhang et al. [27] on the same rig, providing a validated reference point. The addition of the 2.0 mm defect provides a well-developed fault condition where detection is expected to be reliable for both systems, enabling quantification of the baseline SNR gap. The 0.25 mm defect represents a genuinely early-stage fault at the lower boundary of what has been demonstrated in the literature, and is the severity level at which the low-cost measurement chain is most likely to begin losing detection capability. Together, these four levels are designed to reveal the defect size at which performance divergence occurs, directly addressing the primary research question.

For each test, vibration data will be acquired simultaneously from the piezoelectric reference sensor (PCB 352C33) and the MEMS measurement chain, mounted at the same location on the bearing housing. Three MEMS sensors (ADXL1005, ADXL1002, and ADXL335) will be evaluated sequentially across the same fault severity levels, with the piezoelectric reference remaining fixed. As the sensors share a common interface with the ADC, hot swapping between tests should be straightforward, allowing comparison of sensor performance without changing other variables. Each fault severity level will have an independent bearing, and each test will be a relatively short data acquisition period, to adequately capture representative vibration data without inducing significant spall growth beyond the seeded defect size.

The envelope analysis pipeline shown in Section 4.3 will be applied to all signals, with BPFO peak clarity and statistical indicators including RMS and kurtosis used to quantify the performance gap between severity levels and sensor types. Should detection performance prove comparable across all severity levels, the programme will be extended to the two-stage transmission rig to provide a more industrially representative noise environment.

3.4 Resources

The primary experimental resource is the UNSW bearing test rig at Willis Annexe J18, the same rig used by Zhang et al. [27], housing a Nachi 6205-2NSE9 bearing. Due to the floating bearing housing



(a) Bearing test rig (primary testbed).



(b) Two-stage transmission rig (extended testbed).

Figure 8: Experimental testbeds available at UNSW Willis Annexe J18.

design the rig exhibits an exceptionally high signal-to-noise ratio which may not be indicative of real world systems [27]. Therefore, a two-stage transmission rig is also available, presenting a significantly more complex noise environment due to gear mesh interference and electromagnetic interference, and is identified as the intended extended testbed for Thesis B. Both rigs are shown in Fig. 8. The piezo-electric reference sensor is a PCB 352C33 accelerometer. MEMS accelerometers and a NucleoF401RE microcontroller will be purchased (Section 4.2).

3.5 Timeline

Testing will be conducted over ten weeks from 1 June to 7 August 2026, spanning UNSW Term 2, with a detailed Gantt chart provided in Appendix A.1. Component procurement is planned for completion prior to Term 2 to avoid delays. In summary the planned timeline is as follows:

- **Week 1 (1–5 June):** Laboratory training and sensor characterisation, required before experiments can commence
- **Weeks 2–5 (8 June – 5 July):** One week allocated per fault severity level, allowing time for defect milling, sensor mounting, data acquisition, and analysis after each test
- **Weeks 6–7 (6–19 July):** Full results analysis and investigation of MEMS hardware characteristics, dependent on bearing rig outcomes
- **Weeks 8–9 (20 July – 2 August):** Conditional two-stage rig testing, once primary bearing rig results completed.
- **Week 10 (3–7 August):** Buffer for repeat testing or contingencies.

Term 3 (14 September – 20 November 2026) will focus on completion of any remaining experimental work from Term 2, full analysis and interpretation of results, and writing of the final thesis report. If required, further testing on the two-stage transmission rig will be conducted in Term 3, as two weeks in Term 2 is unlikely to be sufficient to fully characterise the effect of more complex noise.

4 Project Dependent Preparations

This section documents the preparatory work completed to date in support of the experimental programme outlined in Section 3. This includes laboratory induction, procurement planning, preliminary signal analysis, and risk assessment.

4.1 Laboratory Induction and Training

Laboratory induction to the UNSW Willis Annexe 203 mechanical engineering laboratory was completed on 24 February 2026, during which the available bearing test rigs were briefly inspected. Formal training on the operation of the bearing test rigs is scheduled for the commencement of Term 2 as part of Thesis B preparations.

4.2 Components and Budget

The primary components to be procured are three MEMS accelerometers of different performance tiers and a microcontroller for data acquisition. The ADXL1002 and ADXL1005 represent high-performance MEMS accelerometers, offering bandwidths of 21 kHz and 23 kHz respectively at a cost of approximately \$40 and \$55 per unit. The ADXL335 represents the low-cost sensor at \$5, with a bandwidth of 0.55 kHz, which may be suitable for tests exploiting the sensor’s own structural resonance. A STM32-based NucleoF401RE microcontroller will also be procured as the data acquisition platform, at a cost of approximately \$20. A summary of planned components and estimated costs is provided in Table 7.

Table 7: Planned component procurement and estimated budget.

Component	Supplier	Unit Cost (USD)	Quantity	Total (USD)
ADXL1002 MEMS	Analog Devices	\$40	1	~\$40
ADXL1005 MEMS	Analog Devices	\$55	1	~\$55
ADXL335 MEMS	Analog Devices	\$5	1	~\$5
NucleoF401RE Microcontroller	STMicroelectronics	\$20	1	\$20
			Total	~\$120

4.3 Preliminary Signal Analysis

To validate the envelope analysis pipeline prior to experimental data collection, a MATLAB script was developed by the author to implement the demodulation-based fault detection methodology described in Section 2.2, and applied to the Mendeley dataset of Zhang et al.[28][27], comprising vibration data using the Nachi 6205-2NSE9 bearing.

Test 1 was selected, corresponding to an outer race fault with a 1.0 mm seeded defect at 10.5 kN radial load [27]. A single measurement file at 20 Hz shaft speed was analysed, with the theoretical BPFO calculated as 71.75 Hz using the bearing geometry reported in [27].

The raw frequency spectrum was inspected to identify a structural resonance band at 7,100–8,400 Hz, which was selected as the demodulation carrier. The envelope was extracted via a manual Hilbert transform implemented in MATLAB using the Fast Fourier Transform, and the squared envelope spectrum computed. The results are presented in Fig. 9, with clear peaks observed at $1\times$, $2\times$, and $3\times$ BPFO, in close agreement with the theoretical value. The MATLAB script used to generate these results is provided in Appendix A.3.

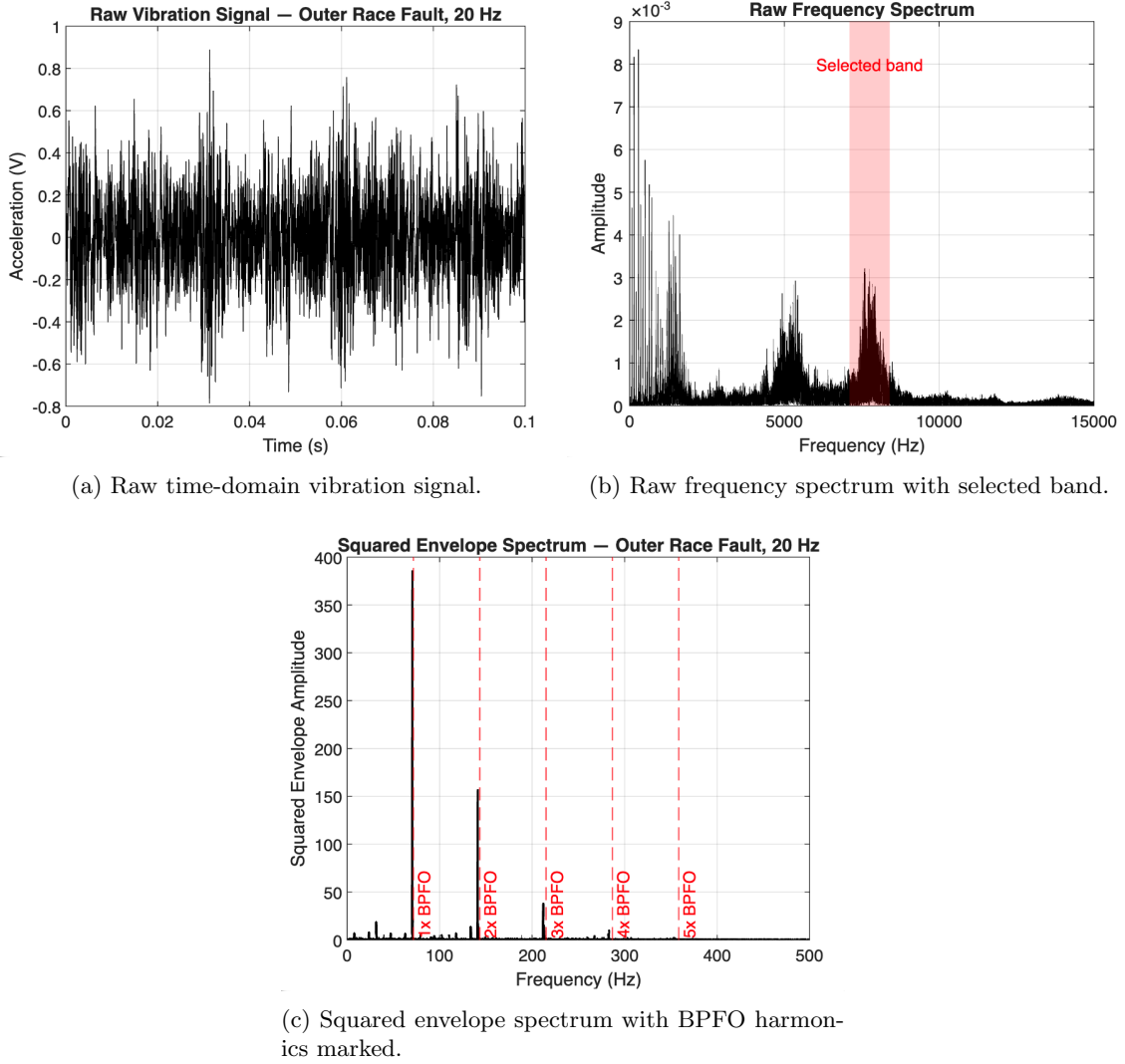


Figure 9: Envelope analysis of Test 1 from Mendeley Data Set [28]. Nachi 6205-2NSE9, outer race fault, 1.0 mm defect, 10.5 kN radial load, 20 Hz. Author's own analysis.

With the pipeline validated on a known dataset, the same methodology will be applied to raw experimental data acquired from the bearing test rig in Thesis B, providing a consistent basis for comparison between the piezoelectric and MEMS measurement chains.

4.4 Risk Assessment

The following risk assessment identifies key risks for successful completion of this thesis, along with their likelihood, potential impact, and mitigation strategies. The key risks identified for this project are summarised in Table 8.

Table 8: Project risk assessment for the experimental programme.

Risk	Likelihood	Consequence	Risk Rating	Mitigation
Bearing failure during testing	Possible	Major	High	Have spare bearings in advance, conduct short acquisition periods only
Component delivery delays	Likely	Major	Very High	Order components before Term 2 starts, identify alternative suppliers
Milling machine unavailability for defect seeding	Unlikely	Major	Medium	Book milling machine in advance, and mill all defects in one go
Two-stage rig EM interference preventing usable data	Likely	Moderate	Medium	Treat as stretch goal
Insufficient sampling rate on NucleoF401RE for high-frequency analysis	Likely	Moderate	Medium	Use aliased envelope analysis/sample low frequency resonance from sensor
Rig access or scheduling conflicts	Unlikely	Moderate	Medium	Book rig time in advance and maintain communication with lab technician

A separate safety risk assessment has been completed in Appendix A.4.

5 Conclusion

This report has established the technical foundation and experimental plan for an investigation into the feasibility of low-cost vibration-based bearing fault detection using MEMS accelerometers. The literature review first established the physical basis of rolling element bearing faults, before examining the signal processing techniques required to extract fault information from measured vibration signals, and finally reviewing the sensing and data acquisition hardware requirements that govern detection capability.

The review identified that while low-cost MEMS measurement chains have been demonstrated to detect bearing faults under favourable conditions [22], the trade-off between cost and detection capability across the full measurement chain remains insufficiently characterised, particularly for early-stage faults in the sub-millimetre severity regime. Two hardware constraints were identified as potentially exploitable for low-cost systems: exploitation of the sensor’s own structural resonance as a demodulation carrier, and envelope analysis of aliased high-frequency content, both representing significant gaps in the context of low-cost MEMS-based bearing fault detection.

To address these gaps, an experimental programme has been designed using the UNSW bearing test rig, evaluating a piezoelectric reference sensor (PCB 352C33) alongside three MEMS sensors (ADXL1005, ADXL1002, and ADXL335) evaluated sequentially, across four outer race fault severity levels of 0.25, 0.5, 1.0, and 2.0 mm. This will directly quantify the performance gap between the systems as a function of fault severity, identify the primary governing factors, and evaluate the viability of the two hardware characteristics identified in the literature review. Preliminary signal analysis using the Zhang et al. [28] Mendeley dataset has validated the envelope analysis pipeline in MATLAB, with clear BPFO peaks identified at the correct characteristic frequency, confirming readiness for experimental data collection in Thesis B.

Acknowledgments

A special thank you to supervisor Dr Pietro Borghesani for his continued guidance and support throughout this project, including thorough review of draft material, regular meetings, and invaluable direction in shaping the research question and experimental approach.

It is also acknowledged that the use of several tools supported the research and writing process. Notion was used for literature management, organisation of research notes, and tracking the mapping of content across sections of this report. Large language models including Claude (Anthropic) and ChatGPT (OpenAI) were used during the early stages of this work to assist with initial ideation, identifying relevant literature, understanding unfamiliar concepts, and improving the clarity and grammar of written material. Technical content, analysis, and conclusions presented in this report are the work of the author.

References

- [1] Alhussein Albarbar, Samir Mekid, Andrew Starr, and Robert Pietruszkiewicz. Suitability of mems accelerometers for condition monitoring: An experimental study. *Sensors*, 8(2):784–799, 2008.
- [2] Analog Devices. Adxl1001/adxl1002: Low noise, high frequency mems accelerometers data sheet, 2018. Accessed: 2026-03-29.
- [3] Analog Devices. Adxl1005: Low noise, wide bandwidth mems accelerometer data sheet, 2018. Accessed: 2026-03-29.
- [4] ATO Inc. Piezoelectric accelerometer sensor (5g–100g), 2025. Accessed: 2026-03-19.
- [5] Gary Foss, Jessica Meloy, Mark Valentino, and Patrick Walter. Sensors and their signal conditioning for dynamic acceleration, force, pressure, and sound applications. In Randall Allemang and Peter Avitabile, editors, *Handbook of Experimental Structural Dynamics*, pages 47–76. Springer, 2011.
- [6] Jingbo Gai, Junxian Shen, Yifan Hu, and He Wang. An integrated method based on hybrid grey wolf optimizer improved variational mode decomposition and deep neural network for fault diagnosis of rolling bearing. *Measurement*, 2020.
- [7] Mohammed Hakim, Abdoulhdi A. Borhana Omran, Ali Najah Ahmed, Muhannad Al-Waily, and Abdallah Abdellatif. A systematic review of rolling bearing fault diagnoses based on deep learning and transfer learning: Taxonomy, overview, application, open challenges, weaknesses and recommendations. *Ain Shams Engineering Journal*, 14(4):101945, 2023.
- [8] L. Magadán, C. Ruiz-Cárcel, J. C. Granda, F. J. Suárez, and A. Starr. Explainable and interpretable bearing fault classification and diagnosis under limited data. *Advanced Engineering Informatics*, 56:102013, 2023.
- [9] Nuno Maia, António Urgueira, Raquel Almeida, and Tiago Silva. Experimental methods for measuring rotational degrees of freedom. In *Toward Robust Response Models: Theoretical and Experimental Issues*, chapter 3.2, pages 1166–1183. Springer, 2011.
- [10] P. D. McFadden and J. D. Smith. Model for the vibration produced by a single point defect in a rolling element bearing. *Journal of Sound and Vibration*, 96(1):69–82, 1984.
- [11] P. D. McFadden and J. D. Smith. Vibration monitoring of rolling element bearings by the high-frequency resonance technique — a review. *Tribology International*, 17(1):3–10, 1984.
- [12] National Instruments. *NI-9234 Specifications*, April 2026. 4-Channel, 51.2 kS/s/channel, ± 5 V, C Series Sound and Vibration Input Module.
- [13] Bin Pang, Guiji Tang, Tian Tian, and Chong Zhou. Rolling bearing fault diagnosis based on an improved htt transform. *Sensors*, 18(4):1209, 2018.
- [14] PCB Piezotronics. *Model 352C33 High Sensitivity Ceramic Shear ICP® Accelerometer Datasheet*. PCB Piezotronics, 2024.

- [15] Robert B. Randall and Jérôme Antoni. Rolling element bearing diagnostics—a tutorial. *Mechanical Systems and Signal Processing*, 25(2):485–520, 2011.
- [16] A. Rossi, G. Bocchetta, F. Botta, and A. Scorza. Accuracy characterization of a mems accelerometer for vibration monitoring in a rotating framework. *Applied Sciences*, 13(8):5070, 2023.
- [17] Jose E. Ruiz-Sarrio, Jose A. Antonino-Daviu, and Claudia Martis. Localized bearing fault analysis for different induction machine start-up modes via vibration time-frequency envelope spectrum. *Sensors*, 24:6935, 2024.
- [18] Stefan Johann Rupitsch. *Piezoelectric Sensors and Actuators: Fundamentals and Applications*. Springer, 2019.
- [19] Omear Saeed. MME applied mechanics lab 203 (general area, A, B, C) general RMF. Technical Report ENG-MECH-RMF-16279, School of Mechanical and Manufacturing Engineering, UNSW Sydney, 2023.
- [20] N. Sawalhi and R. B. Randall. Simulating gear and bearing interactions in the presence of faults part ii: Simulation of the vibrations produced by extended bearing faults. *Mechanical Systems and Signal Processing*, 22(8):1952–1966, 2008.
- [21] Nader Sawalhi and Robert B. Randall. Simulating gear and bearing interactions in the presence of faults. part i: The combined gear bearing dynamic model and the simulation of localised bearing faults. *Mechanical Systems and Signal Processing*, 22(8):1924–1951, 2008.
- [22] Gagandeep Sharma, Tejbir Kaur, Sanjay Kumar Mangal, Nishant Kumar Dhiman, and Gopal Lal Jat. Mems approach for rolling bearing fault diagnosis using vibration signal analysis. *Journal of Vibration Engineering & Technologies*, 13(10), 2025.
- [23] Wade A. Smith, Pietro Borghesani, Qing Ni, Kesheng Wang, and Zhongxiao Peng. Optimal demodulation-band selection for envelope-based diagnostics: A comparative study of traditional and novel tools. *Mechanical Systems and Signal Processing*, 134:106303, 2019.
- [24] César Ricardo Soto-Ocampo, José Manuel Mera, Juan David Cano-Moreno, and José Luis Garcia-Bernardo. Low-cost, high-frequency, data acquisition system for condition monitoring of rotating machinery through vibration analysis—case study. *Sensors*, 20(12):3493, 2020.
- [25] STMicroelectronics. Stm32f401xd and stm32f401xe: Arm cortex-m4 32-bit mcu datasheet, 2013. Accessed: 2026-03-29.
- [26] Navid Yazdi, Farrokh Ayazi, and Khalil Najafi. Micromachined inertial sensors. *Proceedings of the IEEE*, 86(8):1640–1659, 1998.
- [27] Hengcheng Zhang, Pietro Borghesani, Robert B. Randall, and Zhongxiao Peng. A benchmark of measurement approaches to track the natural evolution of spall severity in rolling element bearings. *Mechanical Systems and Signal Processing*, 166:108466, 2022.
- [28] Hengcheng Zhang, Pietro Borghesani, Siyuan Zhuang, and Zhongxiao Peng. Bearing run-to-failure datasets of UNSW, 2021.

A Appendix

A.1 Gantt Chart

insertGantt Chart Here

A.2 Draft Thesis Outline

The following represents the current draft outline of the research thesis, reflecting the structure developed to date. This will be expanded and refined in Thesis B as experimental results are obtained.

Contents

1	Introduction	2
2	Literature Review	3
2.1	Bearing Faults and Diagnostics	4
2.1.1	Types of rolling element bearing faults	4
2.1.2	Characteristic Bearing Fault Frequencies	5
2.2	Signal Processing for Bearing Fault Detection	6
2.2.1	Detection Challenges and Signal Characteristics	6
2.2.2	Demodulation-Based Fault Detection	7
2.3	Sensors and Measurement Chain	8
2.3.1	Piezoelectric Accelerometers	9
2.3.2	MEMS Accelerometers	10
2.3.3	Analog to Digital Converters and Data Acquisition	11
2.3.4	Performance Comparison	12
2.3.5	Detection Capabilities	14
3	Research Question and Project Plan	15
3.1	Research Question and Hypothesis	15
3.2	Aims	15
3.3	Experimental Methodology	16
3.4	Resources	16
3.5	Timeline	17
4	Project Dependent Preparations	18
4.1	Laboratory Induction and Training	18
4.2	Components and Budget	18
4.3	Preliminary Signal Analysis	18
4.4	Risk Assessment	20
5	Conclusion	21
A	Appendix	24
A.1	Gantt Chart	24
A.2	Draft Thesis Outline	26
A.3	MATLAB Code	27
A.4	Safety Risk Assessment	29

A.3 MATLAB Code

```

25 %% Bearing Parameters - NACHI 6205
26 % Outer race fault frequency
27 nb = 9; % Number of balls
28 Bd = 7.94; % Ball diameter
29 Pd = 39.04; % Pitch diameter
30 alpha = 0; % Contact angle
31 fr = 20; % Shaft speed
32
33 BPF0 = (nb/2) * fr * (1 - (Bd/Pd) * cosd(alpha));
34 fprintf('Shaft speed: %.1f Hz\n', fr);
35 fprintf('BPF0: %.2f Hz\n', BPF0);
36
37 %% Plot 1 - Raw Time Domain
38 figure(1);
39 plot(t, x, 'k', 'LineWidth', 0.5);
40 xlabel('Time (s)');
41 ylabel('Acceleration (V)');
42 title('Raw Vibration Signal - Outer Race Fault, 20 Hz');
43 xlim([0 0.1]);
44 grid on;
45 set(gcf, 'Color', 'w');
46
47 %% Filter around resonance
48 X = fft(x);
49 f_vec = (0:N-1) * (fs/N);
50
51 figure(2);
52 plot(f_vec(1:floor(N/2)), abs(X(1:floor(N/2)))/N, 'k', 'LineWidth', 0.5);
53 xlabel('Frequency (Hz)');
54 ylabel('Amplitude');
55 title('Raw Frequency Spectrum');
56 xlim([0 fs/2]);
57 grid on;
58 set(gcf, 'Color', 'w');
59
60 xlim([0 15000]);
61 hold on;
62 patch([7100 8400 8400 7100], [0 0 9e-3 9e-3], 'r', ...
63 'FaceAlpha', 0.2, 'EdgeColor', 'none');
64 text(7750, 8e-3, 'Selected band', 'HorizontalAlignment', 'center', ...
65 'Color', 'r', 'FontSize', 10);
66 hold off;
67

```

Figure 10: MATLAB script (Part 1). Author's own work.

```

68 %% Envelope Analysis
69 fl = 7100;
70 fh = 8400;
71
72 x = x(:);
73 X = fft(x);
74 N = length(x);
75 f_vec2 = (0:N-1)' * (fs/N);
76
77 mask = (f_vec2 >= fl & f_vec2 <= fh) | ...
78        (f_vec2 >= fs-fh & f_vec2 <= fs-fl);
79
80
81 Xbp = X .* mask;
82 x_bp = real(ifft(Xbp));
83
84 Xh = fft(x_bp);
85 h = zeros(N, 1);
86 h(1) = 1;
87 h(2:N/2) = 2;
88 h(N/2+1) = 1;
89 analytic = ifft(Xh .* h);
90 envelope = abs(analytic);
91
92 SES = (abs(fft(envelope)).^2) / N;
93 f_ses = (0:N-1)' * (fs/N);
94
95 fprintf('x_bp RMS: %.6f\n', rms(x_bp));
96 fprintf('envelope RMS: %.6f\n', rms(envelope));
97 figure(4);
98 plot(real(x_bp(1:5000)));
99 title('Bandpassed Signal');

```

```

100 %% Plot 3 - Squared Envelope Spectrum
101
102 SES(1) = 0;
103
104 figure(3);
105 plot(f_ses(1:floor(N/2)), SES(1:floor(N/2)), 'k', 'LineWidth', 1.5);
106 xlabel('Frequency (Hz)');
107 ylabel('Squared Envelope Amplitude');
108 title('Squared Envelope Spectrum - Outer Race Fault, 20 Hz');
109 xlim([0 500]);
110 grid on;
111 set(gcf, 'Color', 'w');
112
113 hold on;
114 for h = 1:5
115     xline(h * BPF0, '--', sprintf('%dx BPF0', h), ...
116           'Color', [1 0 0 0.4], ...
117           'LineWidth', 0.8, ...
118           'LabelVerticalAlignment', 'bottom');
119 end
120 hold off;

```

Figure 11: MATLAB script (Part 2). Author's own work.

A.4 Safety Risk Assessment

Table 9: Safety risk assessment for experimental work in UNSW Applied Mechanics Laboratory J18 [19], in accordance with ENG-MECH-RMF-16279.

Hazard	Likelihood	Consequence	Risk Rating	Mitigation
Rotating machinery (bearing test rig)	Unlikely	Moderate	Medium	Maintain safe distance, never reach into operating rig, complete formal rig training prior to use, ensure guards are in place
Electrical shock (sensors, DAQ, wiring)	Rare	Major	Medium	Inspect all wiring and confirm test tags before use, do not modify live circuits
Trip hazard (cables, equipment)	Unlikely	Minor	Low	Use pendant power outlets, cover floor cables with cable protectors, keep work area clear
Manual handling (moving equipment)	Unlikely	Moderate	Medium	Use correct lifting technique, request assistance for heavy items
Electromagnetic interference (two-stage rig)	Likely	Minor	Low	Use shielded cables, maintain separation between power and signal lines